

Q1 - 24 June - Shift 1

Choose the correct option from the following options given below :

Space for your notes:

- (A) In the ground state of Rutherford's model electrons are in stable equilibrium. While in Thomson's model electrons always experience a net-force.
- (B) An atom has a nearly continuous mass distribution in a Rutherford's model but has a highly non-uniform mass distribution in Thomson's model
- (C) A classical atom based on Rutherford's model is doomed to collapse.
- (D) The positively charged part of the atom possesses most of the mass in Rutherford's model but not in Thomson's model.

Q2 - 24 June - Shift 2

In Bohr's atomic model of hydrogen, let K , P and E are the kinetic energy, potential energy and total energy of the electron respectively. Choose the correct option when the electron undergoes transitions to a higher level :

Space for your notes:

- (A) All K , P and E increase.
- (B) K decreases. P and E increase.
- (C) P decreases. K and E increase.
- (D) K increases. P and E decrease.

Q3 - 25 June - Shift 1

Questions

MathonGo

The ratio for the speed of the electron in the 3rd orbit of He⁺ to the speed of the electron in the 3rd orbit of hydrogen atom will be :-

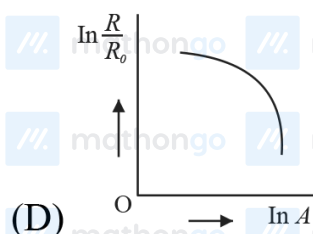
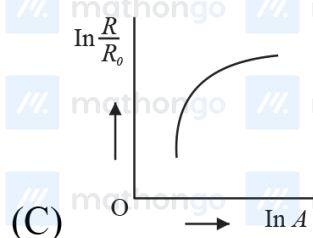
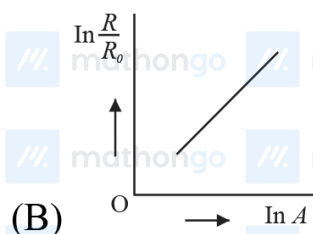
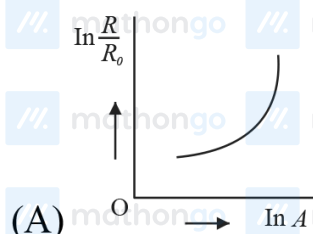
- (A) 1 : 1 (B) 1 : 2
(C) 4 : 1 (D) 2 : 1

Space for your notes:

Q4 - 25 June - Shift 2

Which of the following figure represents the variation of $\ln\left(\frac{R}{R_0}\right)$ with $\ln A$ (If R = radius of a nucleus and A = its mass number)

Space for your notes:



Q5 - 27 June - Shift 1

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Questions

MathonGo

A hydrogen atom in its ground state absorbs 10.2 eV of energy. The angular momentum of electron of the hydrogen atom will increase by the value of : (Given, Planck's constant = 6.6×10^{-34} Js)

(A) 2.10×10^{-34} Js (B) 1.05×10^{-34} Js

(C) 3.15×10^{-34} Js (D) 4.2×10^{-34} Js

*Space for your notes:***Q6 - 27 June - Shift 1**

A beam of monochromatic light is used to excite the electron in Li^{++} from the first orbit to the third orbit. The wavelength of monochromatic light is found to be $x \times 10^{-10}$ m. The value of x is _____.

[Given $hc = 1242$ eV nm]

*Space for your notes:***Q7 - 29 June - Shift 1**

$\sqrt{d_1}$ and $\sqrt{d_2}$ are the impact parameters corresponding to scattering angles 60° and 90° respectively, when an α particle is approaching a gold nucleus. For $d_1 = x d_2$, the value of x will be _____.

Space for your notes:

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Answer Key

Q1 (C)

Q2 (B)

Q3 (D)

Q4 (B)

Q5 (B)

Q6 (114)

Q7 (3)

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Q1 (C)

According to Rutherford, e^- revolves around nucleus in circular orbit. Thus e^- is always accelerating (centripetal acceleration). An accelerating charge emits EM radiation and thus e^- should lose energy and finally should collapse in the nucleus.

Q2 (B)

Based on theory

Q3 (D)

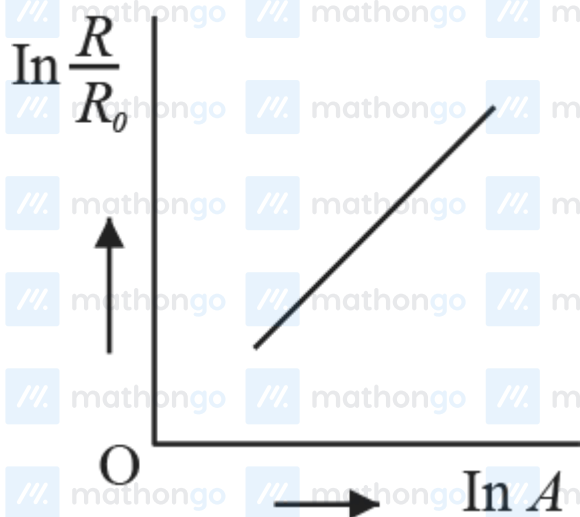
$$v \propto \frac{Z}{n} \propto Z \quad (n = \text{constant})$$

$$\Rightarrow \frac{v_{\text{He}^+}}{v_{\text{H}}} = \frac{Z_{\text{He}^+}}{Z_{\text{H}}} = \frac{2}{1}$$

Q4 (B)

$$R = R_0 A^{\frac{1}{3}}$$

$$\ln \frac{R}{R_0} = \frac{1}{3} \ln A$$



Q5 (B)

$$13.6 \left(\frac{1}{1^2} - \frac{1}{n^2} \right) = 10.2$$

$$n = 2$$

$$L_i = \frac{h}{2\pi} \times 1$$

$$L_F = \frac{2h}{2\pi}$$

$$\Delta L = L_F - L_i = \frac{h}{2\pi} = \frac{6.6 \times 10^{-34}}{2 \times \frac{22}{7}}$$

$$= 1.05 \times 10^{-34} \text{ J-s}$$

Q6 (114)

$$Z = 3$$

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$n_1 = 1, n_2 = 3,$$

$$\frac{1}{\lambda} = R(9) \left(\frac{1}{1} - \frac{1}{9} \right) = 8R$$

$$\lambda = \frac{1}{8R} = 114 \times 10^{-10} m$$

Q7 (3)

$$\sqrt{d} \propto \cot \frac{\theta}{2}$$

$$\cot^2 30^\circ = x \cot^2 45^\circ$$

$$3 = x$$